Institute of Technology of Cambodia
Faculty of Electrical Engineering
Génie Informatique et Communication

GICI5CC · CLOUD COMPUTING · WEEK 7

Databases in the Cloud

Data models, relational and nonrelational databases, and the AWS database services.



Heng Rathpisey · Department GIC · Academic year 2025–2026

Recap

LAST SESSION

Network

A common cloud service: networking in the cloud.

LAST SESSION

VPC, security groups, network ACLs

The virtual network on AWS and its two firewalls.

Learning objectives

1. Describe key database concepts.
2. Compare and contrast relational and nonrelational databases.
3. Identify the AWS database services.

THIS WEEK

Database

The nature of data, database fundamentals, and the types of databases available.

PART 1

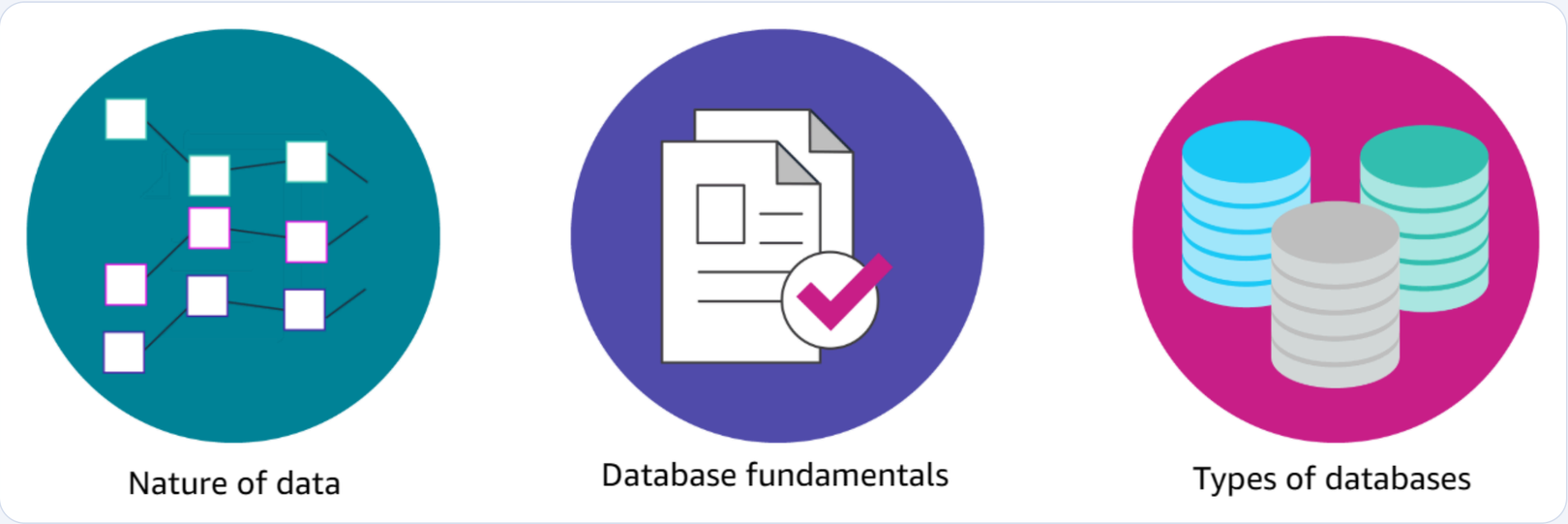
Introduction to databases

The nature of data, database fundamentals, and the types of databases.

Week 7

Introduction to databases

Databases provide a consistent, organized structure for storing and retrieving large amounts of data.



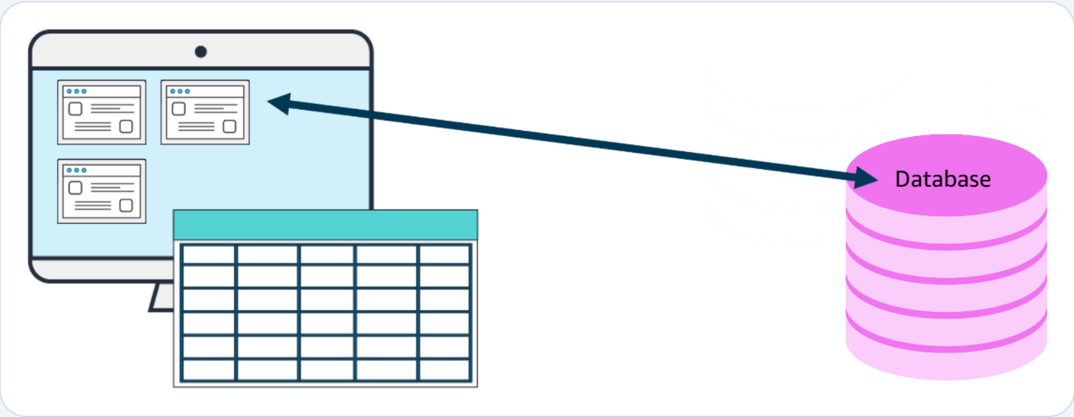
They let you build more complex applications that can take orders, process payments, make recommendations and more.

Why a database?

A database is a *logically organized* collection of information.

A computer needs to store information before it can be referenced or changed, and to find the right information at the right time.

A database is designed so a computer program can access the information later. It gives access to the data while keeping its integrity in a secure environment.

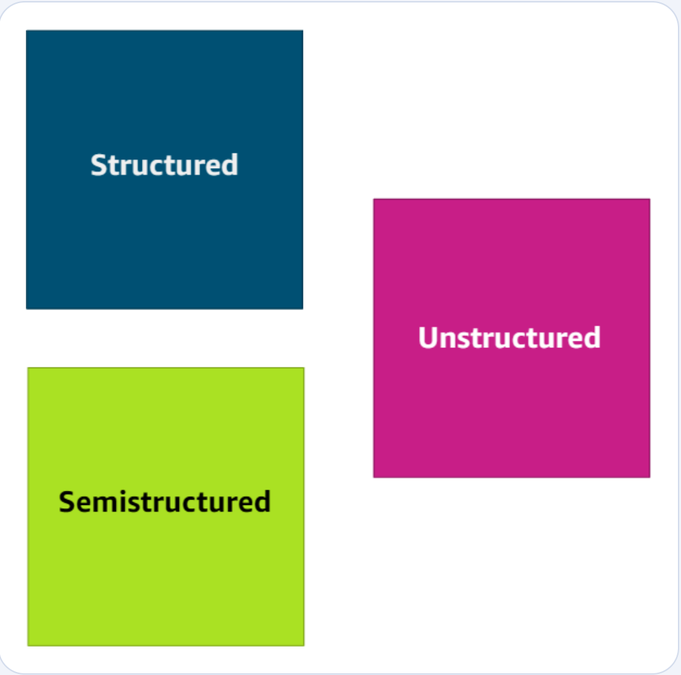


Data models and structure

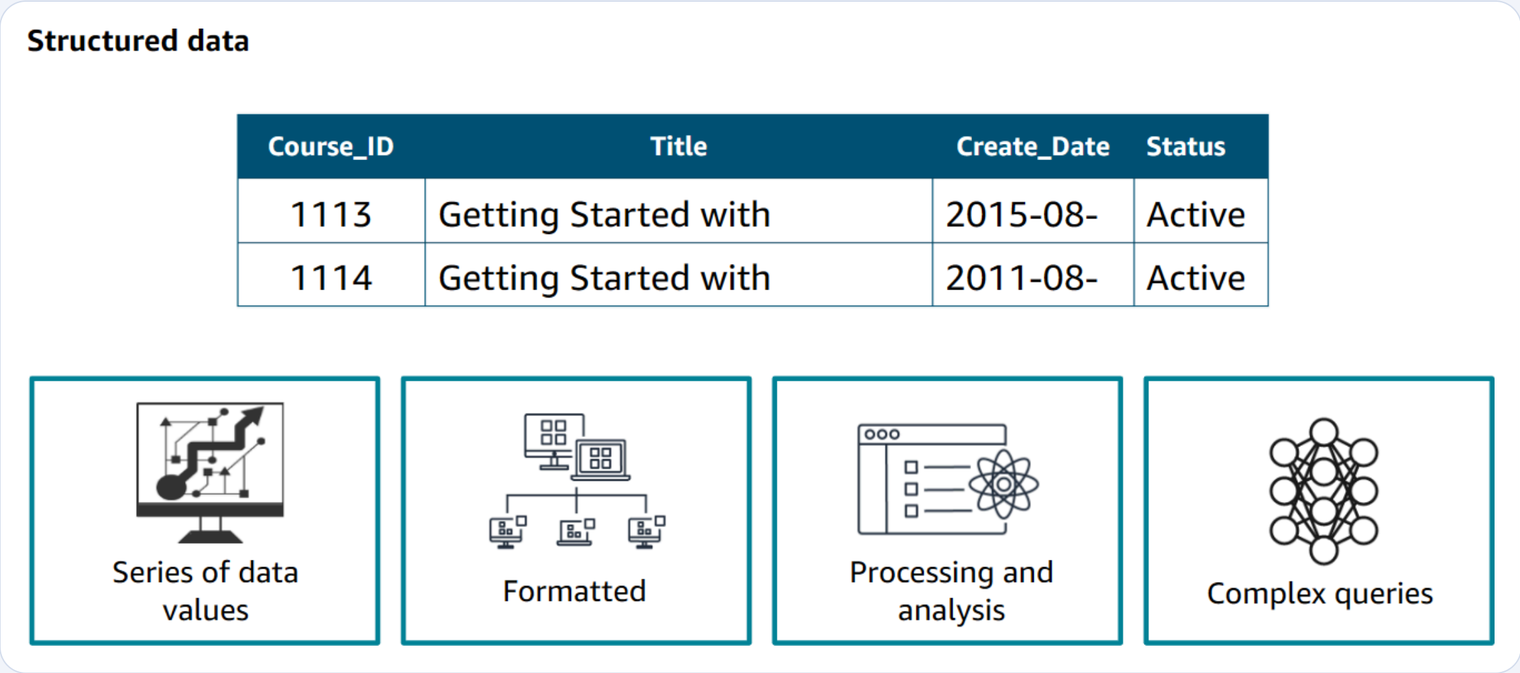
A data model is the *logical structure* of a database.

It determines the rules for how information can be organized and used.

The data model you choose is influenced by the structure of your data. Data is structured in three different ways: structured, unstructured and semistructured.

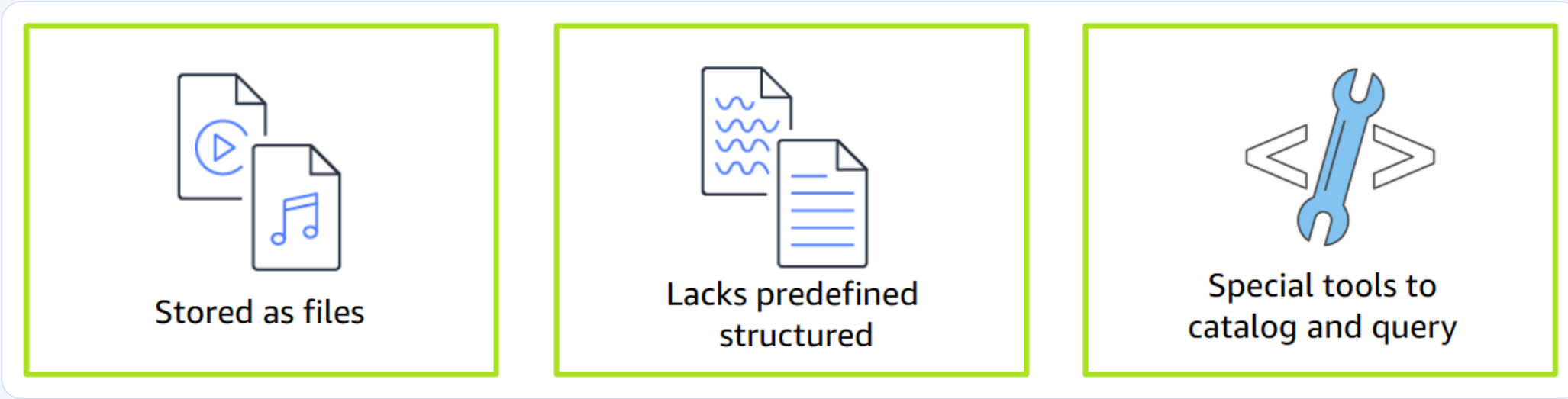


Structured data



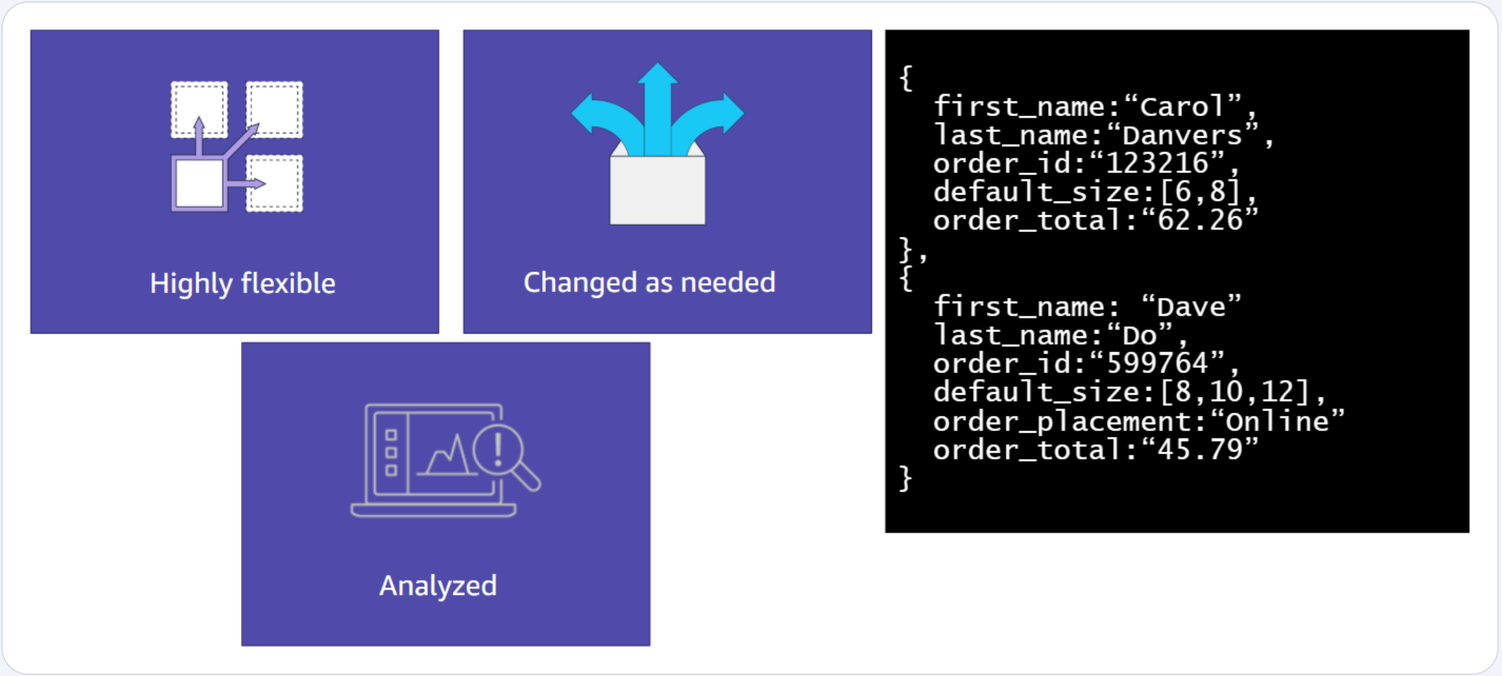
Stored as a series of data values in related tables, so its elements are addressable and it can be used in highly complex queries.

Unstructured data



Stored as files. It lacks any predefined structure and requires special tools to catalog and query the data.

Semistructured data



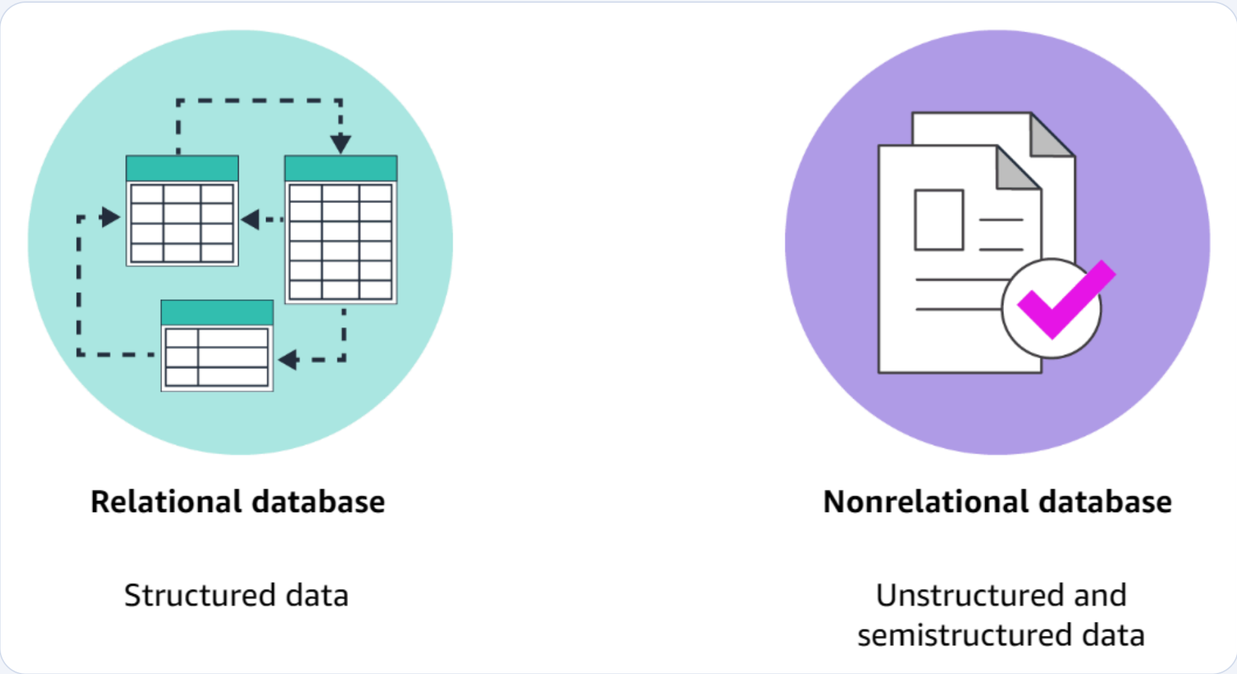
The structure is not strict and can be changed as needed. It can be analyzed, but not with the complexity that structured data allows.

PART 2

Relational and nonrelational

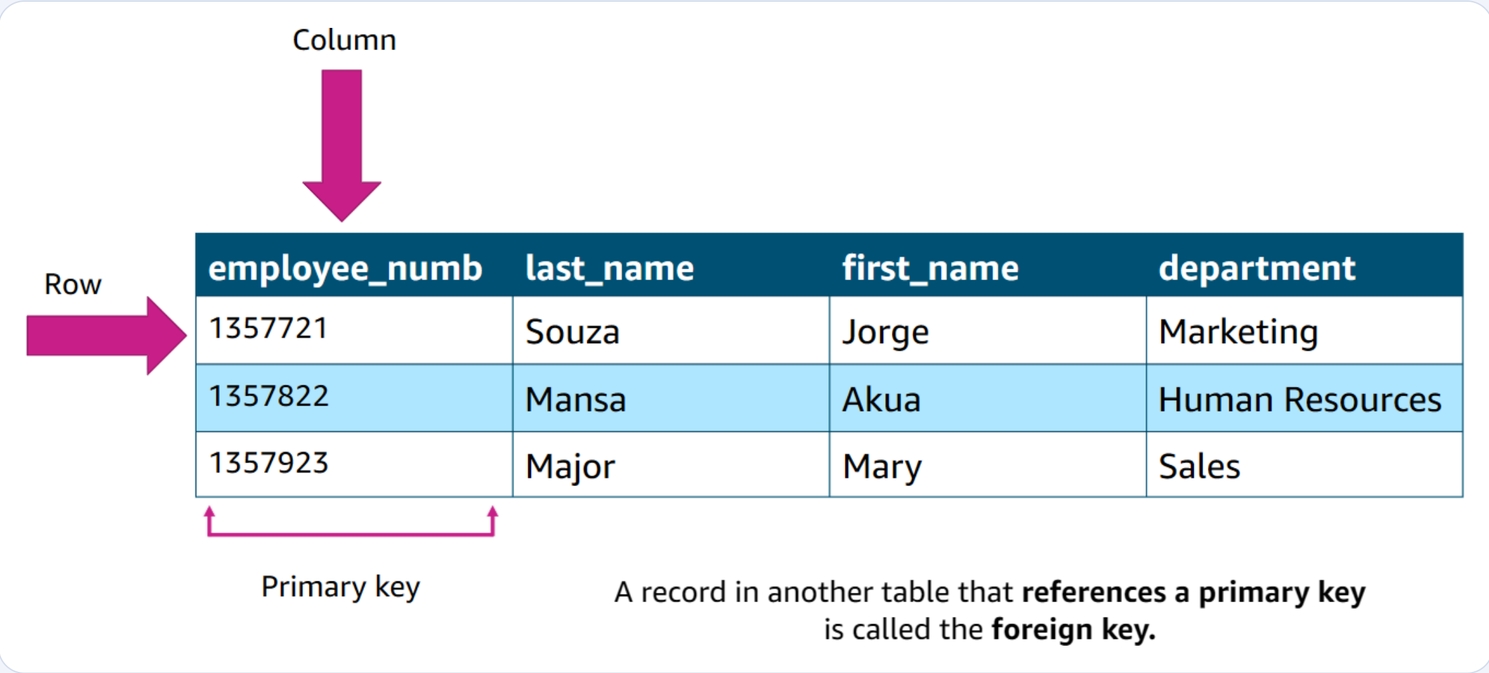
Two families of databases, chosen by the structure of the data.

Relational and nonrelational databases



Relational databases hold structured data. Nonrelational databases hold unstructured and semistructured data.

Relational database tables

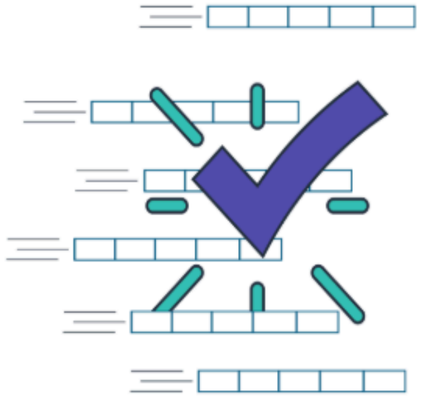


A table has columns and rows. The primary key identifies each row. A record in another table that references it is a foreign key.

Relational database benefits



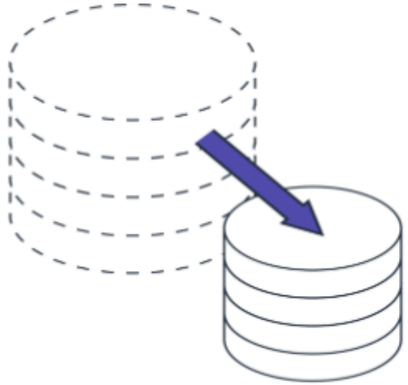
Ease of use for structured data



Data integrity controls and accuracy

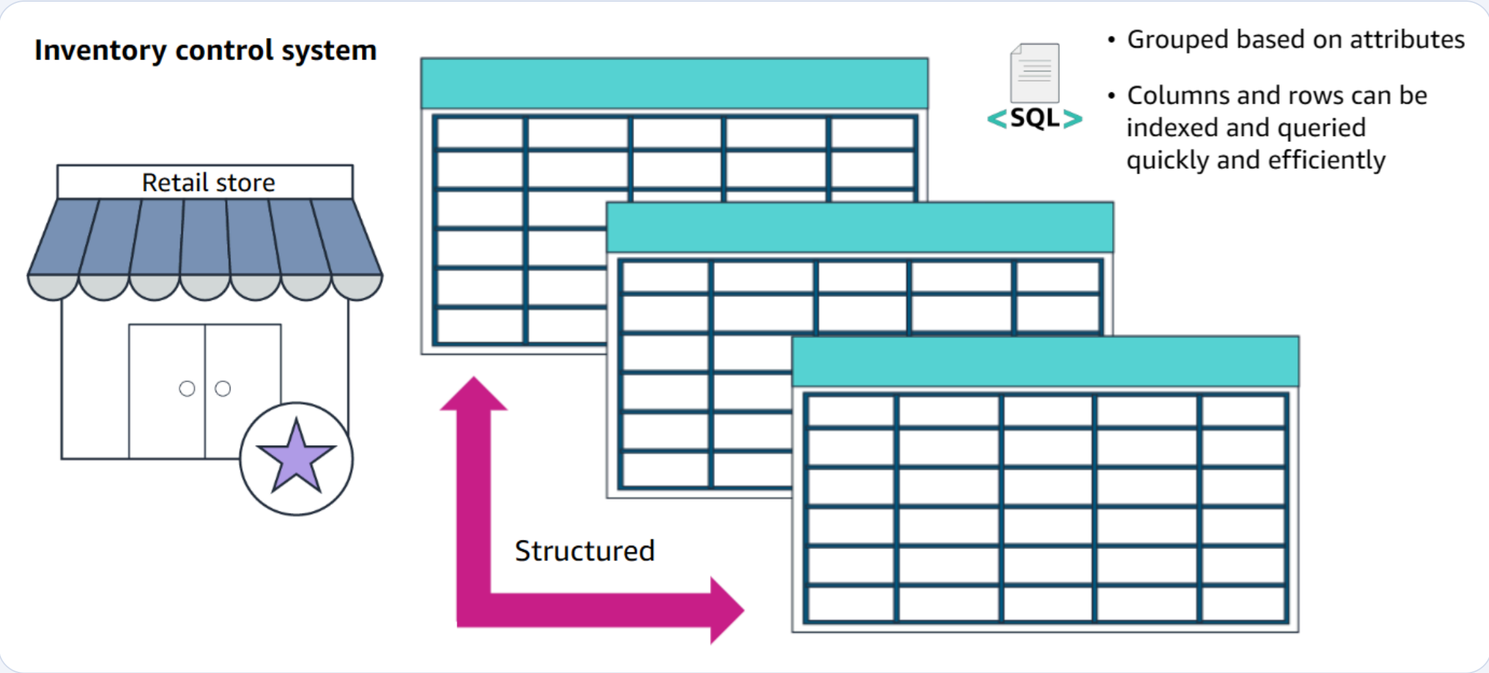


Common shared query language



Reducing redundancy and overall data storage

Relational database use case



An inventory control system: data is grouped by attributes, and columns and rows can be indexed and queried quickly.

Nonrelational databases

A nonrelational database stores data in *unstructured* ways, using one of many storage models.

Storage models

Examples include key-value pairs, documents and graphs.

Schemas are dynamic

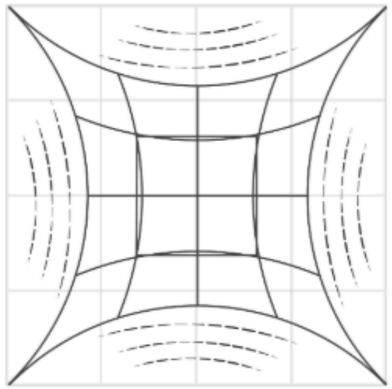
A row does not have to contain data for each column.

Scale horizontally

They scale out, and can work with unstructured and semistructured data.

They grew in popularity because they were designed to overcome the limitations of relational databases for handling the demands of variable structured data.

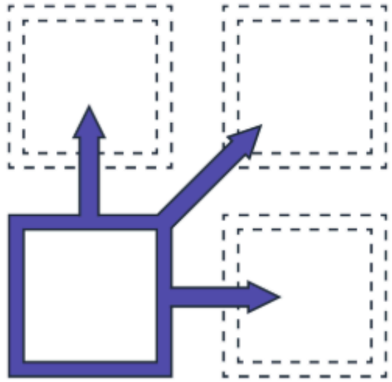
Nonrelational database benefits



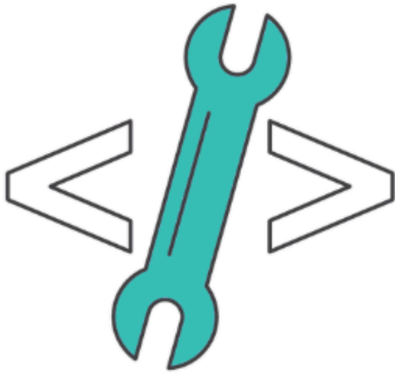
Flexibility



Scalability



High performance



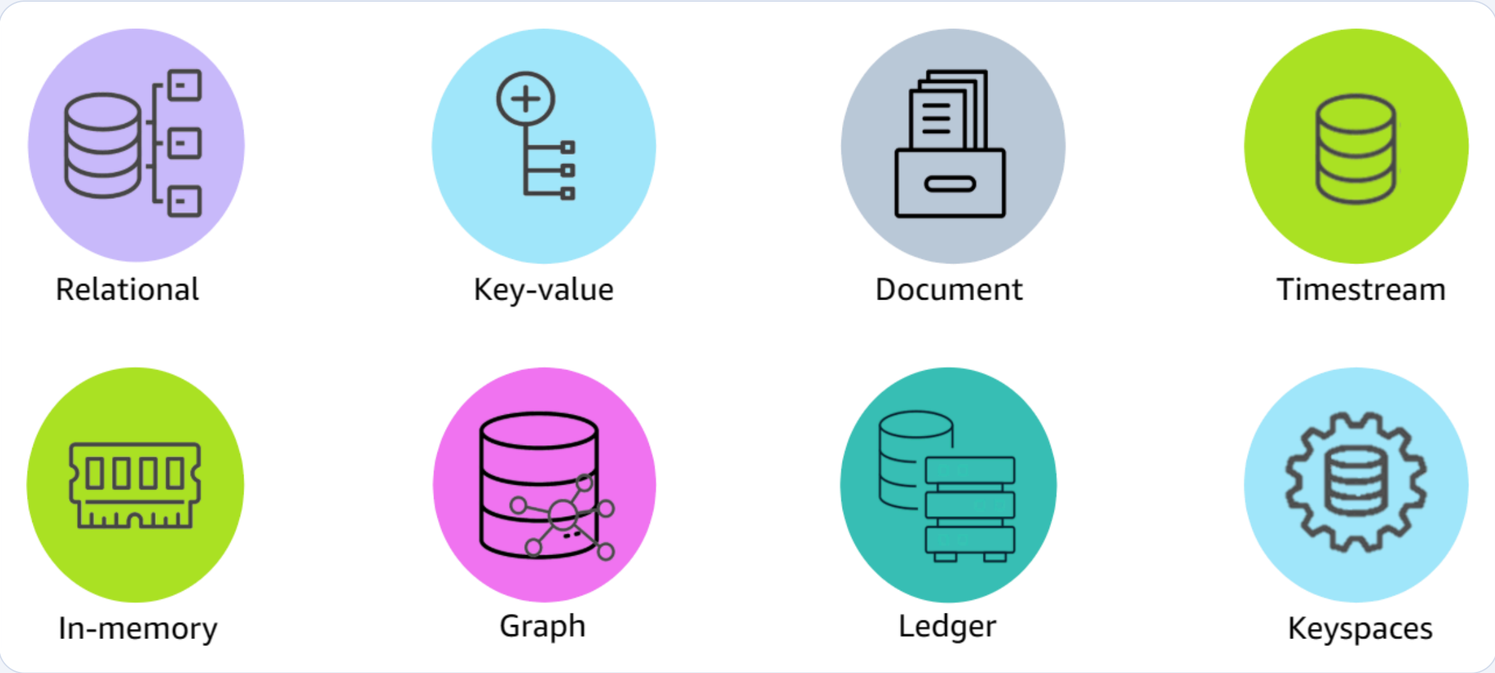
Highly functional APIs

Nonrelational database use case



A personalized customer experience: track customer data in near real time, and take in all types of data from a variety of sources.

AWS databases



AWS has several cloud-native, purpose-built database services, one for each kind of data.

Key takeaways

01

Structured data

Relational databases are best for structured data.

02

Unstructured data

Nonrelational databases are best for unstructured data.

03

Purpose built

AWS has several cloud-native, purpose-built database services.

Next session



01

Serverless